

Developer Technology Trends: Current Usage and Future Demand

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EXECUTIVE SUMMARY



Key findings from job postings, salary data and 18,845 developer survey responses

- Washington DC (5,316), Detroit (3,945) and Seattle (3,375) lead job postings; Swift (\$130.8K) and Python (\$114.4K) top the salary ranking
- JavaScript, SQL and TypeScript are both the most used and the most desired languages
 - Rust and Go enter the desired top 10, while PHP and PowerShell drop out
- PostgreSQL is the leading database today (11,514) and its lead grows next year (12,193); Redis rises to 2nd place
- AWS, Microsoft Azure and Google Cloud dominate cloud platforms; React and Node.js dominate web frameworks
- Most respondents are 25-44 years old (68.6%) and hold a bachelor's or master's degree (72.3%)



INTRODUCTION



Purpose of the report, target audience & value of the analysis

- Purpose: identify which technology skills are in demand today and which ones developers want to adopt next
- Target audience: hiring managers, HR and training teams, and job seekers planning their skills
- Value: data-driven guidance on which languages, databases, platforms and frameworks to learn or invest in
- Scope of the analysis:
 - Job postings by US city and salaries by programming language
 - Developer survey: current vs. desired technologies and respondent demographics



METHODOLOGY

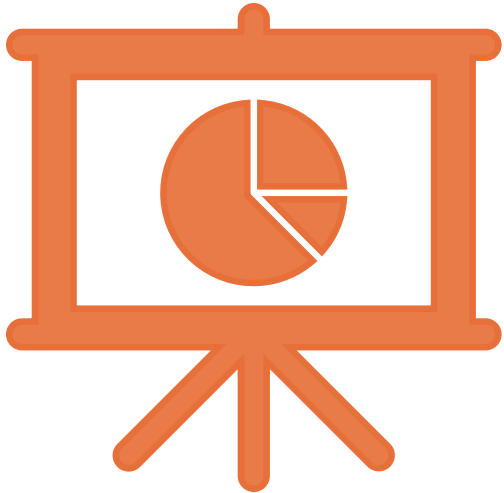


Data sources, data collection & data wrangling steps

- Data sources: job postings via a Job API (job-postings.xlsx), salaries by language via web scraping (popular-languages.csv), Stack Overflow Developer Survey
- Data wrangling in Python (pandas):
 - Removed 10 duplicate rows (65,447 to 65,437 responses)
 - Imputed missing categorical values with the mode: RemoteWork (10,631 rows, "Hybrid") and EdLevel (4,653 rows, "Bachelor's degree")
 - Imputed missing ConvertedCompYearly (64%) with the median (\$65,000), then normalized it with Min-Max and Z-score scaling
 - Detected 978 compensation outliers with the IQR method (above \$220,861)
- Exploratory analysis: distributions, outliers and correlations (pandas, Matplotlib, Seaborn, SQL on SQLite)
- Dashboard: multi-select answers split into a long table (18,845 respondents), null values excluded, built in Google Looker Studio



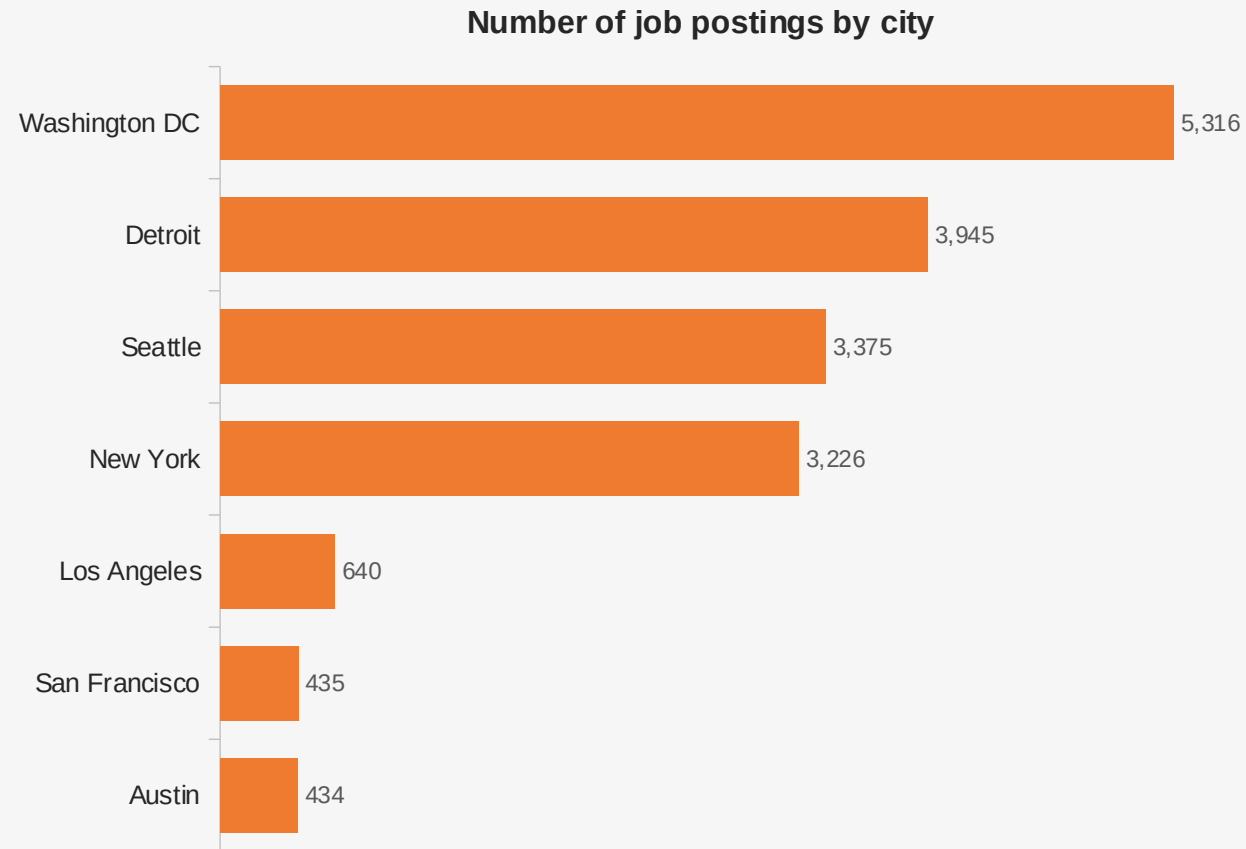
Results



Job postings, salaries by language, and programming language and database trends



JOB POSTINGS



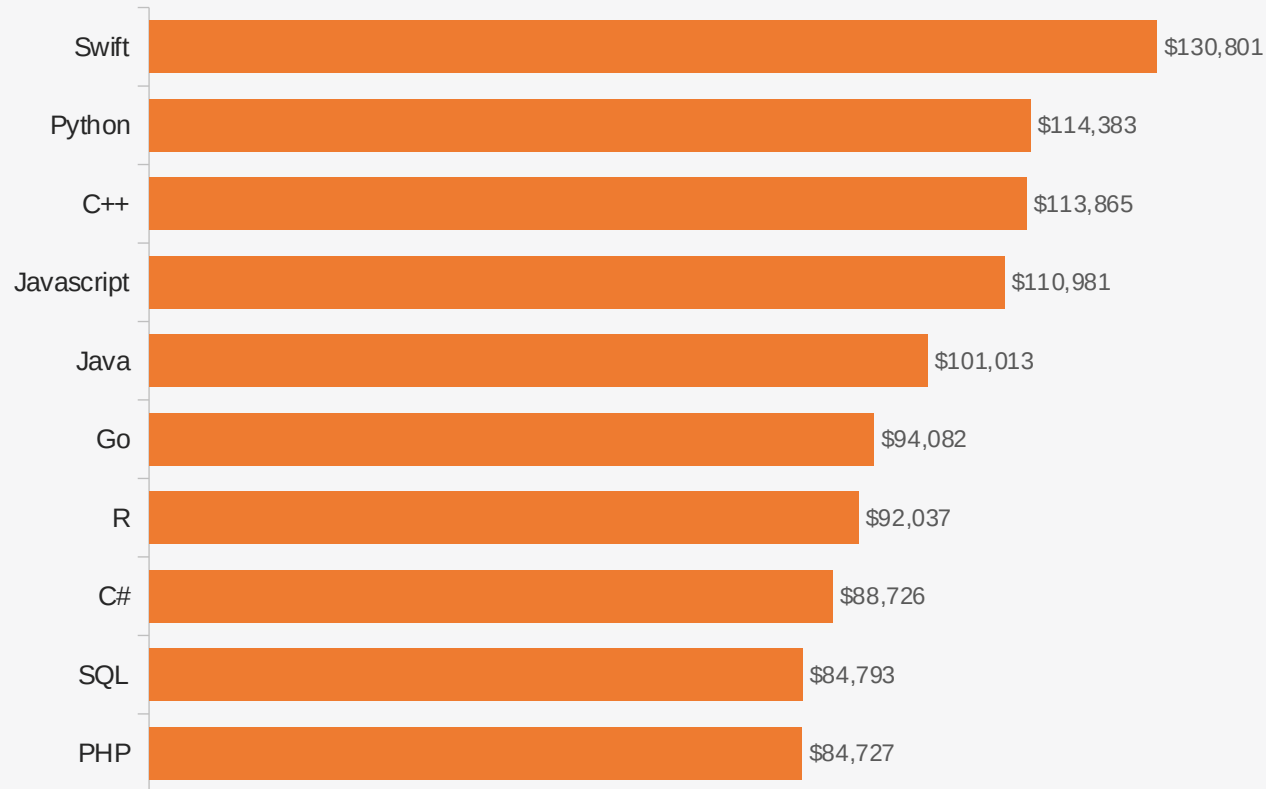
Key takeaways

- Washington DC has the most postings (5,316), ahead of Detroit (3,945) and Seattle (3,375)
- New York follows closely (3,226)
- Los Angeles, San Francisco and Austin have far fewer postings (434-640)



POPULAR LANGUAGES

Average annual salary by programming language



Key takeaways

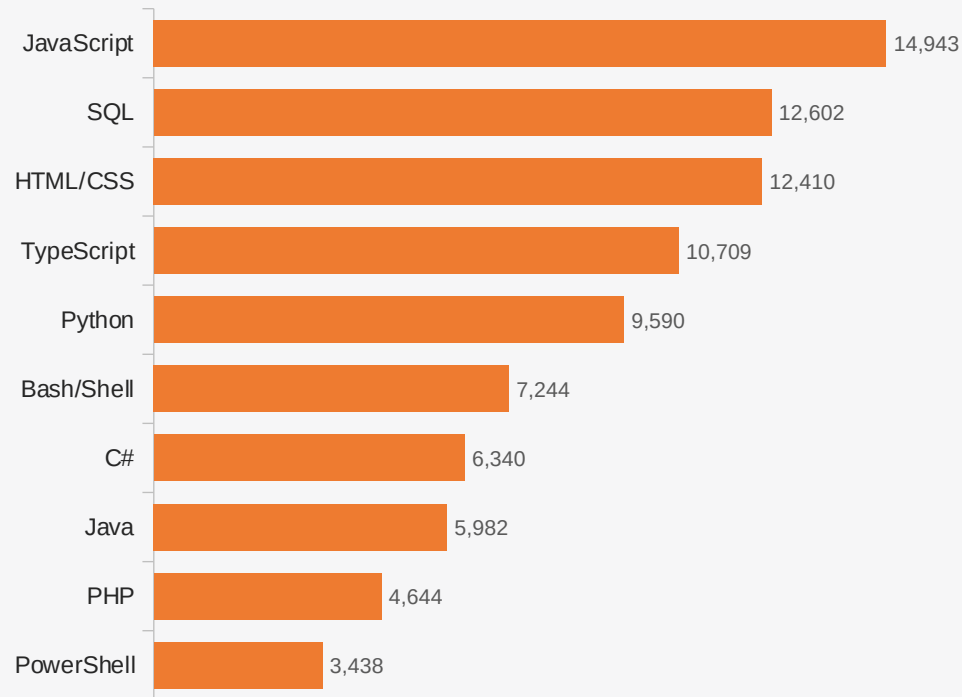
- Swift is the best-paid language (\$130,801)
- Python, C++ and JavaScript all exceed \$110,000
- SQL and PHP are at the bottom (~\$84,700-\$84,800), although SQL is one of the most used languages



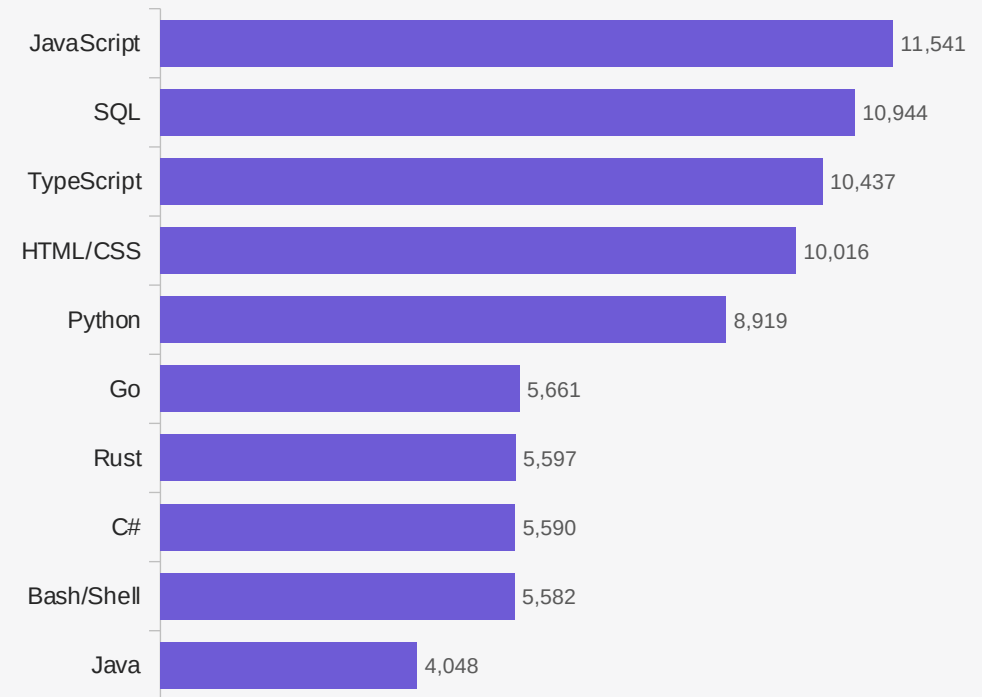
PROGRAMMING LANGUAGE TRENDS

JavaScript, SQL and TypeScript lead both years; Rust and Go replace PHP and PowerShell in next year's top 10.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript is #1 in both years (14,943 used, 11,541 desired)
- TypeScript moves up from 4th to 3rd place, overtaking HTML/CSS
- Rust (5,597) and Go (5,661) enter the desired top 10; PHP and PowerShell leave it
- Python stays 5th in both years

Implications

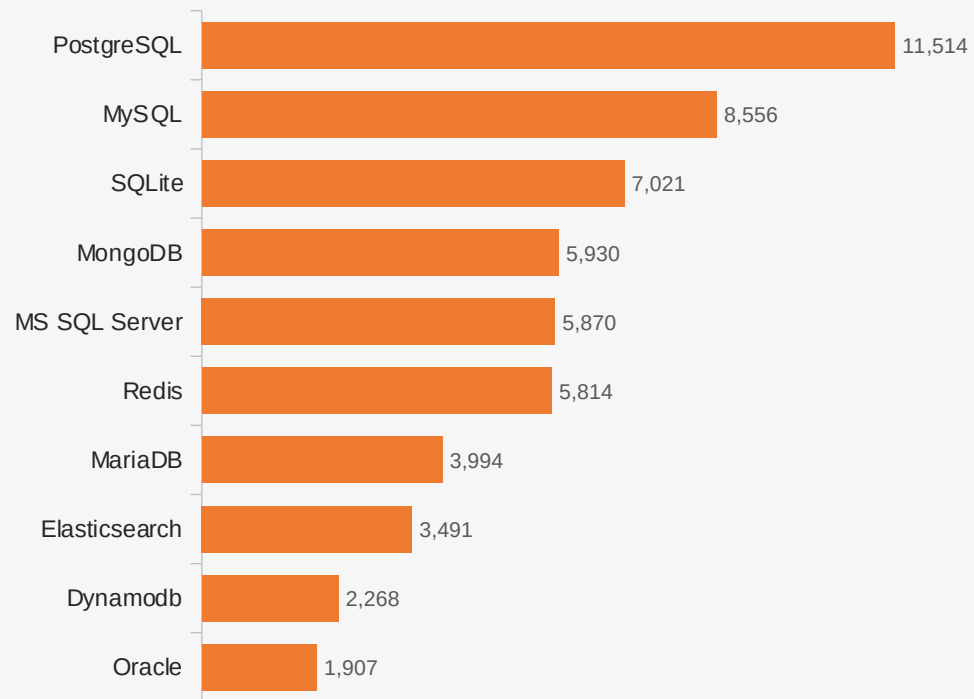
- Web skills (JavaScript/TypeScript) remain a safe core investment
- SQL stays essential for any data-related role
- Training in Rust and Go can anticipate growing demand for systems and cloud development
- Legacy languages (PHP, PowerShell) are losing interest



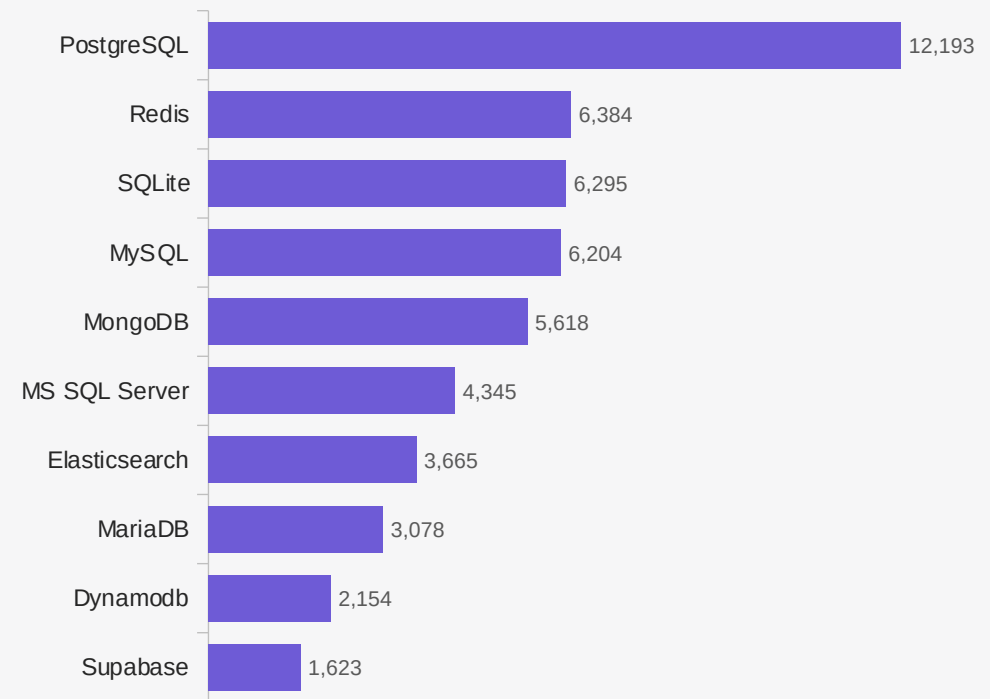
DATABASE TRENDS

PostgreSQL leads both years with a growing gap; Redis jumps to 2nd place and Supabase enters next year's top 10.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is #1 in both years and grows from 11,514 to 12,193
- Redis rises from 6th to 2nd place (6,384)
- MySQL drops from 2nd to 4th (8,556 to 6,204)
- Supabase enters the top 10; Oracle leaves it

Implications

- PostgreSQL is the database skill to prioritize
- Caching and in-memory stores (Redis) are becoming more relevant
- Teams relying on MySQL or Oracle may need migration plans
- Managed, developer-friendly services (Supabase) are gaining interest



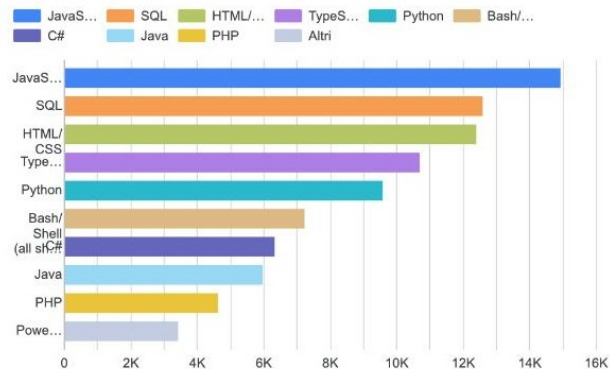
DASHBOARD



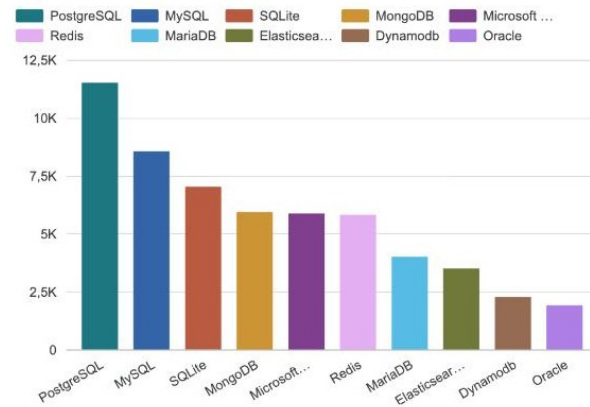
Google Looker Studio dashboard: Current Technology Usage, Future Technology Trends and Demographics

DASHBOARD TAB 1: Current Technology Usage

Top 10 Languages Used



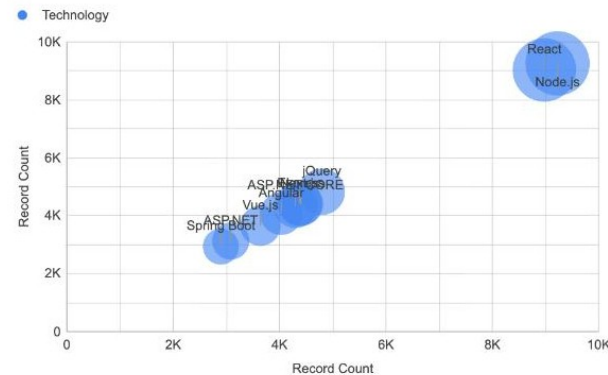
Top 10 Databases Used



Top 10 Platforms Used



Top 10 Web Frameworks Used

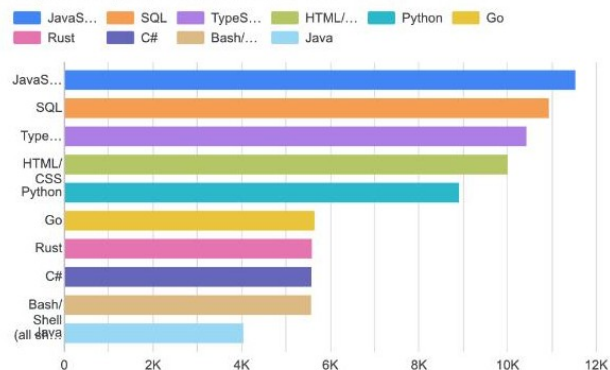


Key insights

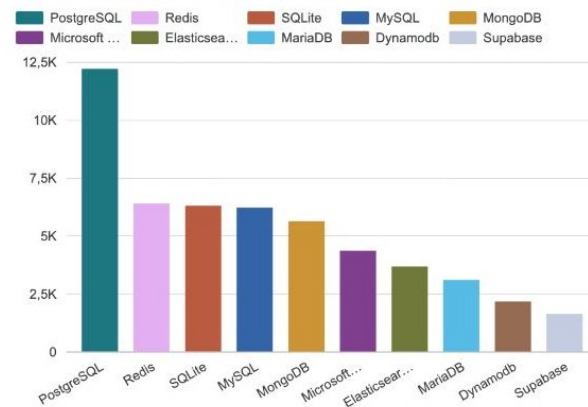
- JavaScript is the most used language (14,943), followed by SQL, HTML/CSS, TypeScript and Python
- PostgreSQL leads the databases (11,514), ahead of MySQL and SQLite
- AWS is the dominant cloud platform (10,871); Microsoft Azure and Google Cloud follow
- Node.js (9,230) and React (8,999) far outpace every other web framework

DASHBOARD TAB 2: Future Technology Trends

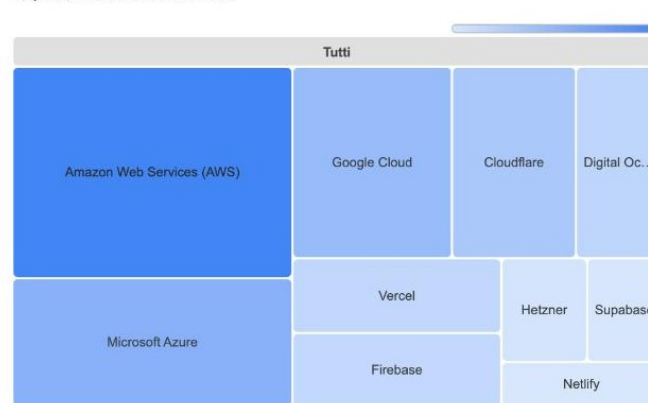
Top 10 Languages Desired Next Year



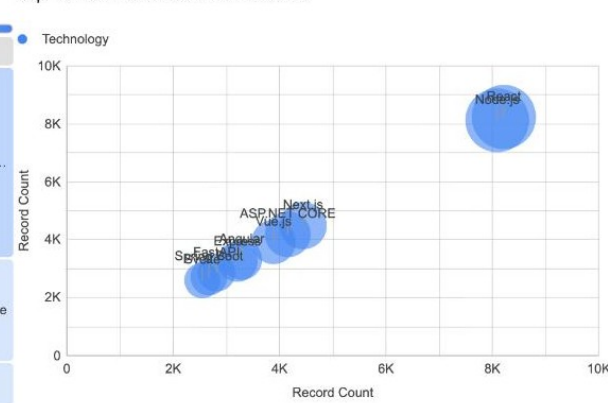
Top 10 Databases Desired Next Year



Top 10 Desired Platforms



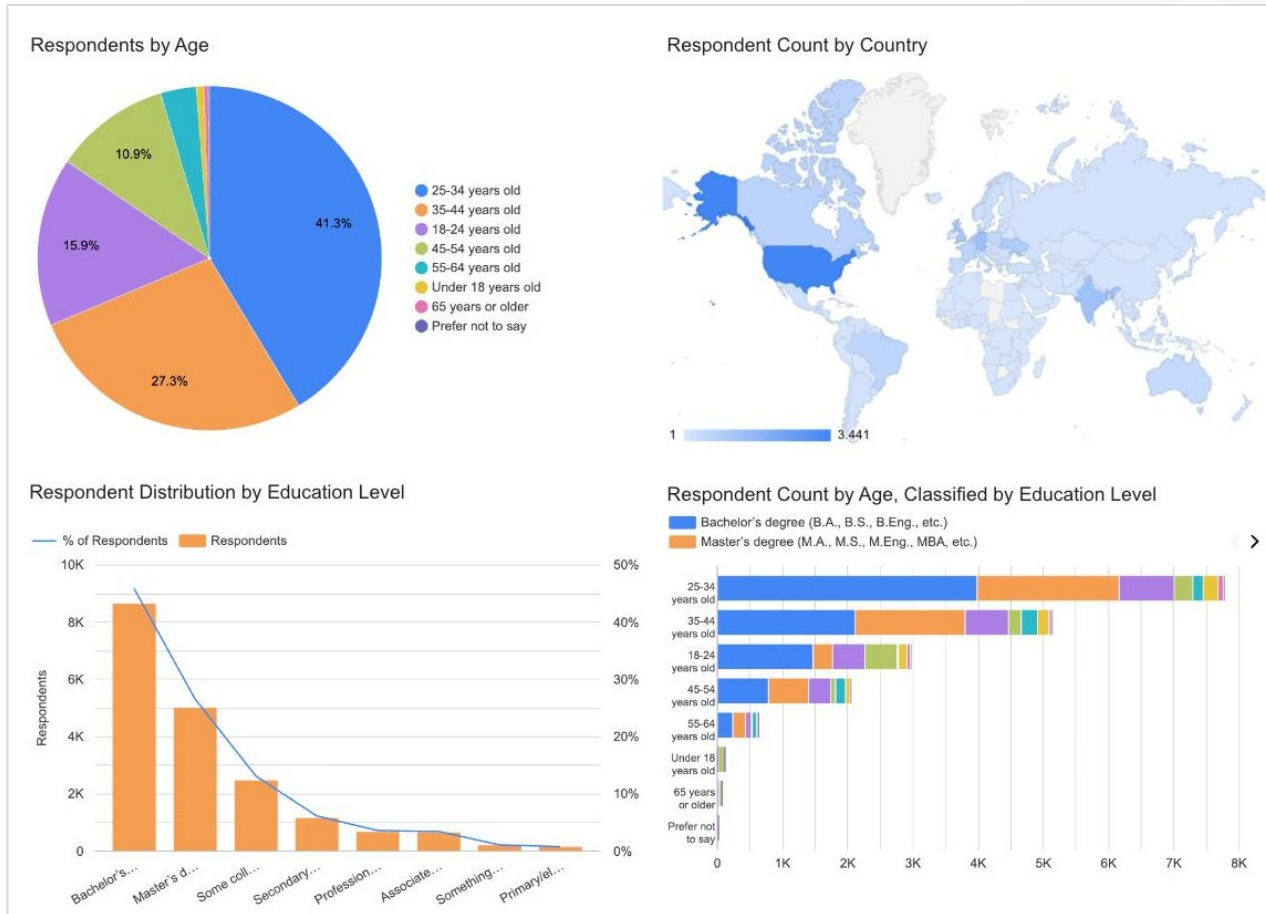
Top 10 Desired Web Frameworks



Future trends

- JavaScript, SQL and TypeScript stay on top; TypeScript overtakes HTML/CSS
- Rust and Go enter the top 10, while PHP and PowerShell drop out
- PostgreSQL widens its lead; Redis rises to 2nd place and Supabase appears
- Cloudflare gains interest; Supabase replaces Heroku among platforms
- React and Node.js remain the most wanted; jQuery leaves the top 10, FastAPI and Svelte enter

DASHBOARD TAB 3: Demographics



Demographic insights

- Respondents are young: 25-34 year olds are 41.3%, and 85% are under 45
- The United States has the most respondents (3,441), followed by Germany (1,339) and India (1,316)
- Bachelor's degree is the most common education level (45.8%), then Master's (26.5%)
- Bachelor's degrees lead in every adult age group; Master's degrees weigh more from age 35 (about 30%)



DISCUSSION



Overall insights derived from the dashboards

- The technology stack is stable at the top: JavaScript, SQL, PostgreSQL, AWS and React/Node.js lead in both current usage and future interest
- Change happens in the middle of the rankings: Rust, Go, Redis, Supabase, FastAPI and Svelte are gaining ground
- Older technologies (PHP, PowerShell, Oracle, jQuery, Heroku) are losing interest
- The survey mainly reflects young, highly educated developers, mostly from the US and Europe, so results may not represent every market

OVERALL FINDINGS & IMPLICATIONS

Findings

- Core skills are stable: JavaScript, SQL, Python, PostgreSQL and AWS stay at the top
- Modern alternatives grow: TypeScript, Rust, Go, Redis and Next.js gain ground
- Demand is concentrated geographically (Washington DC, Detroit, Seattle, New York)
- Salary does not follow popularity: Swift pays the most, SQL and PHP the least

Implications

- Job seekers should combine a solid core (SQL, JavaScript/Python) with one emerging skill
- Companies should plan training on TypeScript, Rust/Go and cloud platforms
- Recruiting efforts can focus on the cities with the most postings
- Niche skills such as Swift can command a salary premium



CONCLUSION



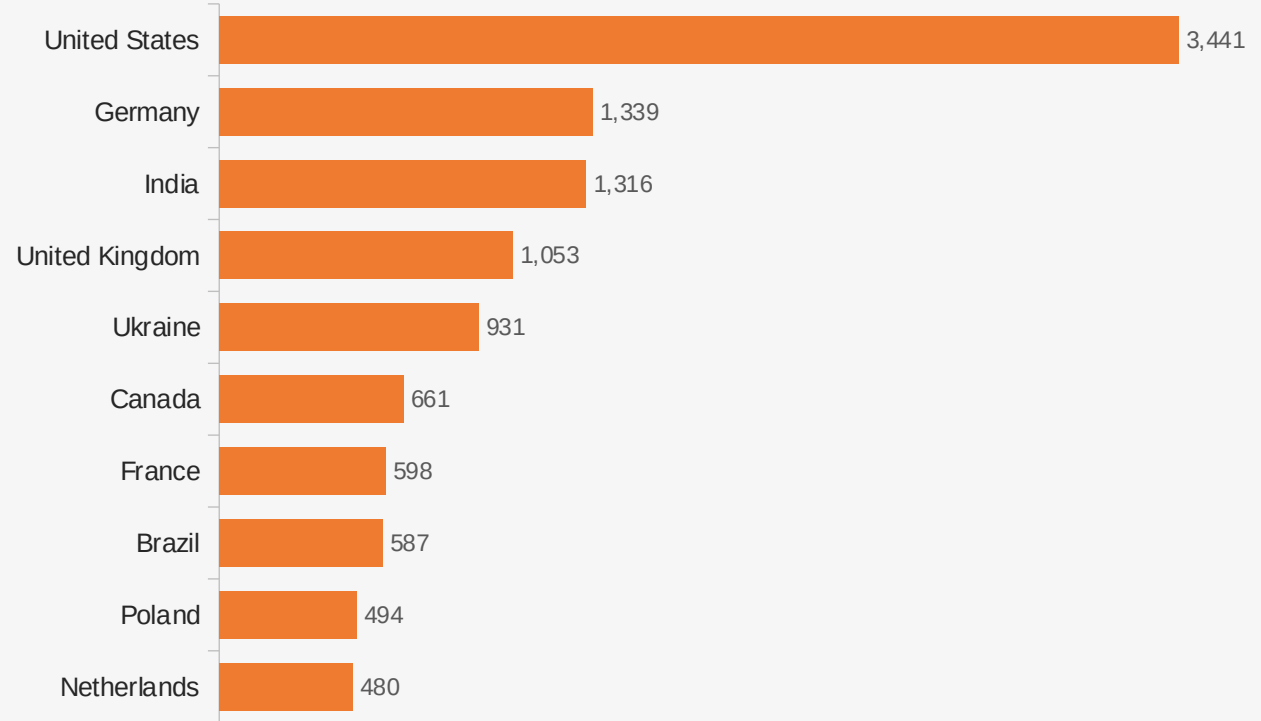
- JavaScript, SQL and PostgreSQL are the most widely used technologies and remain the most desired
- Rust, Go, TypeScript and Redis are the main growing technologies for next year
- AWS, Azure and Google Cloud dominate cloud platforms; React and Node.js dominate web frameworks
- Job opportunities and salaries vary strongly by location and language: monitoring them helps target skills and recruiting



APPENDIX



Top 10 countries by number of survey respondents (161 countries in total)

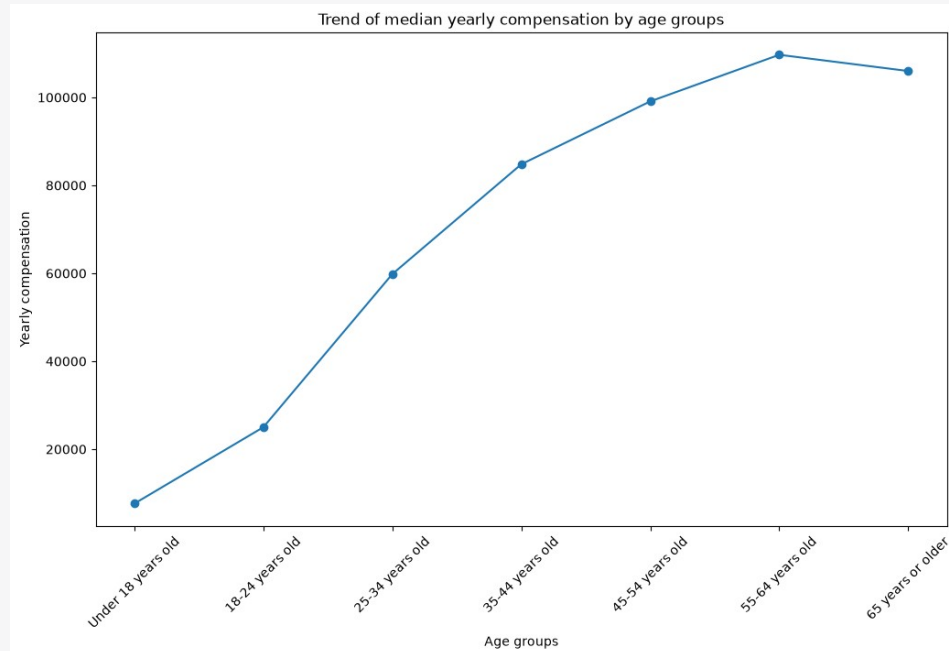


APPENDIX (CONTINUED)



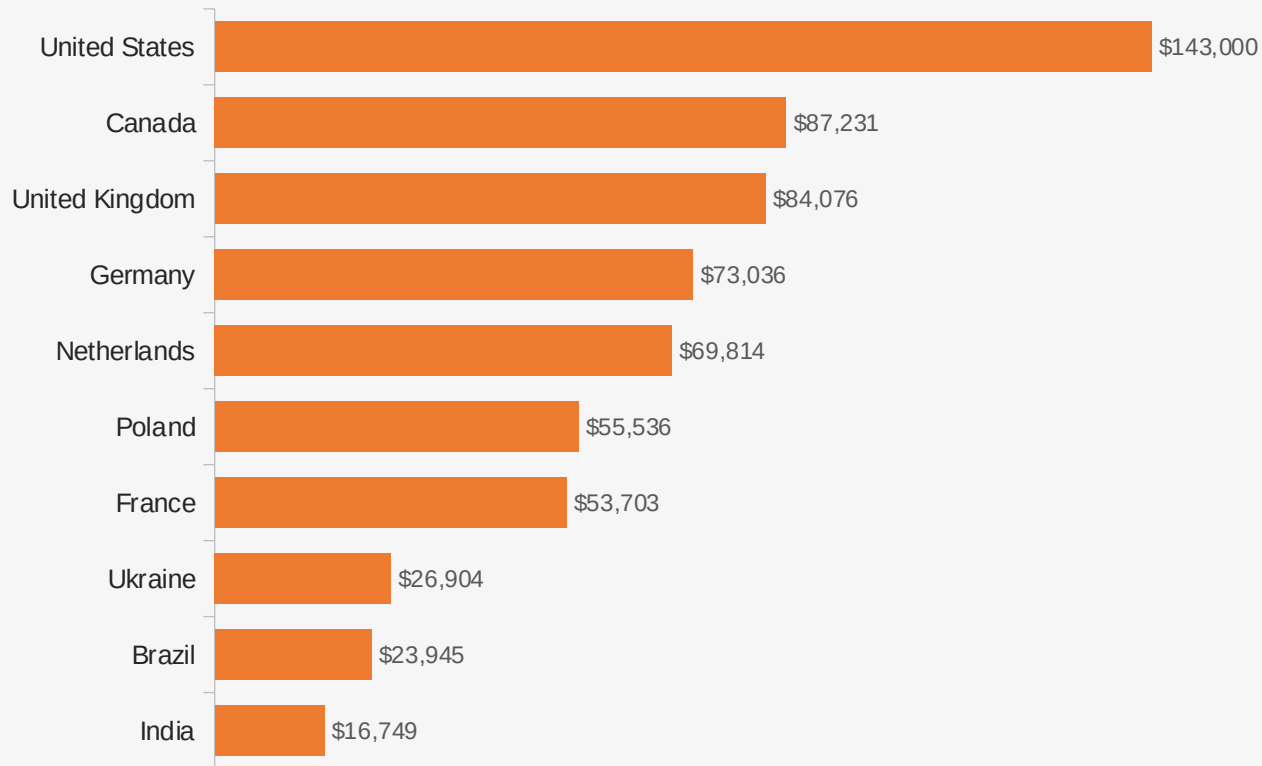
Median yearly compensation by age group (Python, Matplotlib)

- Median pay rises steadily with age, from about \$25K (18-24) to a peak of about \$110K (55-64)



APPENDIX (CONTINUED)

Median yearly compensation (USD) in the top 10 countries by respondents



\$65,000

Overall median yearly compensation (respondents reporting pay)

\$69,814

Median yearly compensation of full-time employees

$r = 0.41$

Correlation between compensation and work experience (outliers removed)

8.5x

Gap between US and India median compensation

Source: Stack Overflow Developer Survey, ConvertedCompYearly (respondents who reported compensation); exploratory analysis in Python (pandas, Seaborn)

